SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I K. Mohammed Yasin *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Ds 8
Problem Understanding	20%	1	2	2	1	2
Method Selection	20%	1	2	1	1	2
Solution Process	25%	2	1	2	2	2
Accuracy of Calculation	20%	1	1	1	1	1
Interpretation of Results	15%	1	1	1	1	1

Total Marks 34 **/40**

Signature of the Faculty

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Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block 172.16.0.0/16 for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address 172.16.10.0/24 and has implemented subnetting using a /27 subnet mask for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address 172.16.10.78 belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address 172.16.10.95 falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address 2001:0db8:0000:0000:0000:ff00:0042:8329. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria / Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding 20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection 20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	Logical,	Mostly correct	Disorganized	No clear process

25%	systematic steps with correct approach throughout	steps with minor mistakes	steps; multiple errors	
Accuracy of Calculation 20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results 15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

CSA07_CO2_AT1_ANALYTICAL PROBLEM SOLVING - Answers

Answer 1: IP Subnetting for Three University Labs

Given:
- Network address: 192.168.20.0/24
- Lab A: /26 subnet
- Lab B: /27 subnet
- Lab C: /28 subnet

We need to find the network address, valid host IP range, broadcast address, and number of usable hosts for each lab.

Step 1: Calculate the subnet details

The formula for usable host IP addresses is:
Usable Hosts = 2^h - 2
Here, h is the number of host bits. We subtract 2 because the network address and broadcast address cannot be assigned to hosts.

Lab	CIDR	Subnet Mask	Host Bits	Total IPs	Usable Hosts
Lab A	/26	255.255.255.192	6	64	62
Lab B	/27	255.255.255.224	5	32	30
Lab C	/28	255.255.255.240	4	16	14

Step 2: Determine the subnet addresses

Lab A — /26
- Network address: 192.168.20.0
- Subnet mask: 255.255.255.192
- Block size: 256 – 192 = 64
- Broadcast address: 192.168.20.63
- Valid host range: 192.168.20.1 – 192.168.20.62
- Usable hosts: 2^6 - 2 = 62

Lab B — /27
The next available subnet begins at 192.168.20.64.
- Network address: 192.168.20.64
- Subnet mask: 255.255.255.224
- Block size: 256 – 224 = 32
- Broadcast address: 192.168.20.95

- Valid host range: 192.168.20.65 – 192.168.20.94
- Usable hosts: 2^5 - 2 = 30

Lab C — /28
The next available subnet begins at 192.168.20.96.
- Network address: 192.168.20.96
- Subnet mask: 255.255.255.240
- Block size: 256 – 240 = 16
- Broadcast address: 192.168.20.111
- Valid host range: 192.168.20.97 – 192.168.20.110
- Usable hosts: 2^4 - 2 = 14

Step 3: Final answer table

Details	Lab A	Lab B	Lab C
Network Address	192.168.20.0	192.168.20.64	192.168.20.96
CIDR Prefix	/26	/27	/28
Subnet Mask	255.255.255.192	255.255.255.224	255.255.255.240
First Valid Host	192.168.20.1	192.168.20.65	192.168.20.97
Last Valid Host	192.168.20.62	192.168.20.94	192.168.20.110
Broadcast Address	192.168.20.63	192.168.20.95	192.168.20.111
Usable Hosts	62	30	14

Conclusion
The network 192.168.20.0/24 is divided into three non-overlapping subnets using different subnet masks. Lab A supports 62 usable hosts, Lab B supports 30 usable hosts, and Lab C supports 14 usable hosts.

Total usable host IP addresses across all three labs: 62 + 30 + 14 = 106

These three subnets occupy 112 IP addresses in total, from 192.168.20.0 to 192.168.20.111, leaving the remaining addresses in the original /24 block available for future subnetting.

Answer 2: Subnetting Analysis for Two Systems
Given:
- System 1 IP address: 192.168.1.130
- System 2 IP address: 192.168.1.190
- Subnet mask: 255.255.255.192

Step 1: Identify the subnet details
The subnet mask 255.255.255.192 corresponds to /26.
Number of host bits: h = 32 - 26 = 6
Number of usable host addresses: 2^6 - 2 = 64 - 2 = 62
Block size: 256 - 192 = 64
Therefore, the subnet ranges in the last octet are 0–63, 64–127, 128–191, and 192–255.

Step 2: Calculate the subnet for each system

Both IP addresses, 130 and 190, fall within the range 128–191. Therefore, both systems belong to the same subnet.

Details	System 1	System 2
IP Address	192.168.1.130	192.168.1.190
Subnet Mask	255.255.255.192	255.255.255.192
CIDR Prefix	/26	/26
Network Address	192.168.1.128	192.168.1.128
First Valid Host	192.168.1.129	192.168.1.129
Last Valid Host	192.168.1.190	192.168.1.190
Broadcast Address	192.168.1.191	192.168.1.191
Usable Host Addresses	62	62

Step 3: Verify whether the IP addresses are valid

- System 1: 192.168.1.130 lies between 192.168.1.129 and 192.168.1.190. Hence, it is a valid host IP address.
- System 2: 192.168.1.190 lies within the same valid host range. Hence, it is also a valid host IP address.

Both systems belong to the same subnet, so they can communicate directly at the IP layer without requiring a router, assuming no VLAN or other network restrictions prevent communication.

Conclusion

The subnet mask 255.255.255.192 divides the network into blocks of 64 IP addresses. Both systems belong to the subnet 192.168.1.128/26, with broadcast address 192.168.1.191 and valid host range 192.168.1.129–192.168.1.190.
Each subnet has 62 usable host IP addresses. Both configured IP addresses are valid host addresses in the same subnet.

Answer 3: Subnetting for Organization Departments

Given:
- IP block: 172.16.0.0/16
- HR: /25
- Sales: /26
- IT: /27
- Admin: /28

We need to calculate the network address, valid host IP range, broadcast address, and usable host addresses for each department.

Step 1: Calculate subnet details

Formula:
Usable Hosts = 2^h - 2
Here, h = 32 - prefix length.

Department	CIDR	Subnet Mask	Host Bits	Total IPs	Usable Hosts
HR	/25	255.255.255.128	7	128	126
Sales	/26	255.255.255.192	6	64	62
IT	/27	255.255.255.224	5	32	30
Admin	/28	255.255.255.240	4	16	14

Step 2: Assign subnet addresses

Allocate the subnets sequentially, starting from 172.16.0.0, ensuring that each subnet starts at a valid boundary.

1. HR Department — /25
- Network address: 172.16.0.0
- Subnet mask: 255.255.255.128
- Valid host range: 172.16.0.1 – 172.16.0.126
- Broadcast address: 172.16.0.127
- Usable hosts: 126

2. Sales Department — /26
The next available address is 172.16.0.128.
- Network address: 172.16.0.128
- Subnet mask: 255.255.255.192
- Valid host range: 172.16.0.129 – 172.16.0.190
- Broadcast address: 172.16.0.191
- Usable hosts: 62

3. IT Department — /27
The next available address is 172.16.0.192.
- Network address: 172.16.0.192
- Subnet mask: 255.255.255.224
- Valid host range: 172.16.0.193 – 172.16.0.222
- Broadcast address: 172.16.0.223
- Usable hosts: 30

4. Admin Department — /28
The next available address is 172.16.0.224.
- Network address: 172.16.0.224

- Subnet mask: 255.255.255.240
- Valid host range: 172.16.0.225 – 172.16.0.238
- Broadcast address: 172.16.0.239
- Usable hosts: 14

Step 3: Final subnet allocation table

Details	HR	Sales	IT	Admin
Network Address	172.16.0.0	172.16.0.128	172.16.0.192	172.16.0.224
CIDR Prefix	/25	/26	/27	/28
First Valid Host	172.16.0.1	172.16.0.129	172.16.0.193	172.16.0.225
Last Valid Host	172.16.0.126	172.16.0.190	172.16.0.222	172.16.0.238
Broadcast Address	172.16.0.127	172.16.0.191	172.16.0.223	172.16.0.239
Usable Hosts	126	62	30	14

Conclusion

The IP block 172.16.0.0/16 is divided into four non-overlapping subnets using the given subnet masks.
- HR supports 126 usable hosts.
- Sales supports 62 usable hosts.
- IT supports 30 usable hosts.
- Admin supports 14 usable hosts.

Total usable host addresses: 126 + 62 + 30 + 14 = 232

The four departments use 256 IP addresses in total, from 172.16.0.0 to 172.16.0.255. The remaining addresses in the /16 block are available for future departmental subnets.

Answer 4: Subnetting Using a /27 Subnet Mask

Given:
- IP address block: 172.16.10.0/24
- New subnet mask: /27 (255.255.255.224)
- IP addresses to check: 172.16.10.78 and 172.16.10.95

Step 1: Calculate the total number of subnets

Formula:
Number of Subnets = 2^n
Here, n is the number of bits borrowed from the host portion.
n = 27 - 24 = 3
Therefore, 2^3 = 8 subnets

Step 2: Calculate usable hosts per subnet

The number of host bits is: h = 32 - 27 = 5
Formula:
Usable Hosts = 2^h - 2
= 2^5 - 2 = 32 - 2 = 30
Therefore, each subnet supports 30 usable host addresses.

Step 3: Determine the subnet of 172.16.10.78

Subnet mask: 255.255.255.224
Block size: 256 - 224 = 32
The subnet ranges in the last octet are:
- 0–31
- 32–63
- 64–95
- 96–127
- 128–159
- 160–191
- 192–223
- 224–255

Since 78 falls between 64 and 95, the IP address 172.16.10.78 belongs to the 64–95 subnet.
- Network address: 172.16.10.64
- First valid host: 172.16.10.65
- Last valid host: 172.16.10.94
- Broadcast address: 172.16.10.95
- Usable hosts: 30

Step 4: Check whether 172.16.10.95 belongs to the same subnet

The IP address 172.16.10.95 is the broadcast address of the subnet 172.16.10.64/27.
Therefore, it belongs to the same subnet range, but it cannot be assigned to an individual host, because it is reserved for broadcasting.

Step 5: Final answer table

Parameter	Answer
Original Network	172.16.10.0/24
New Subnet Mask	255.255.255.224 (/27)
Bits Borrowed	3
Total Subnets	8
Usable Hosts per Subnet	30
Subnet of 172.16.10.78	172.16.10.64/27
Network Address	172.16.10.64
Valid Host Range	172.16.10.65 – 172.16.10.94
Broadcast Address	172.16.10.95
Is 172.16.10.95 in the Same Subnet?	Yes, as the broadcast address
Can 172.16.10.95 Be Assigned to a Host?	No

Conclusion

The /27 subnet mask divides the original /24 network into 8 subnets, each with 30 usable host addresses. The IP address 172.16.10.78 belongs to subnet 172.16.10.64/27, whose broadcast address is 172.16.10.95. Hence, 172.16.10.95 belongs to the same subnet but is reserved as the broadcast address.

Answer 5: IPv6 Address Compression and Network Prefix

Given IPv6 address:
2001:0db8:0000:0000:0000:ff00:0042:8329
Prefix length: /64

Step 1: Compress the IPv6 address

IPv6 compression rules:
1. Leading zeros in each hexadecimal block can be removed.
2. A continuous sequence of zero blocks can be replaced by :: only once in an address.

Original address:
2001:0db8:0000:0000:0000:ff00:0042:8329
Remove leading zeros from each block:
2001:db8:0:0:0:ff00:42:8329
Replace the consecutive zero blocks with ::.
Shortest compressed form:
2001:db8::ff00:42:8329

Step 2: Expand the compressed address

The compressed address is:
2001:db8::ff00:42:8329
An IPv6 address contains 8 hexadecimal blocks. The compressed address contains 5 explicit blocks, so :: represents 3 zero blocks.

Expanded full form:
2001:0db8:0000:0000:0000:ff00:0042:8329
This matches the original IPv6 address.

Step 3: Determine the network prefix using /64

An IPv6 address contains 128 bits.
- Network bits = 64
- Host bits = 128 − 64 = 64
The first 4 hexadecimal blocks represent the 64-bit network prefix.
Network prefix:
2001:db8:0:0::/64
The network prefix in full notation is:
2001:0db8:0000:0000::/64

Step 4: Calculate the total number of host addresses

Formula:

Host Addresses = 2^Host Bits

Since the number of host bits is 64:

2^64 = 18,446,744,073,709,551,616

Therefore, the network contains 2^64 possible interface identifier values.

Step 5: Final answer table

Parameter	Answer
Original IPv6 Address	2001:0db8:0000:0000:0000:ff00:0042:8329
Compressed Address	2001:db8::ff00:42:8329
Expanded Address	2001:0db8:0000:0000:0000:ff00:0042:8329
Prefix Length	/64
Network Prefix	2001:db8:0:0::/64
Network Bits	64
Host Bits	64
Total Possible Host Values	2^64
Total in Decimal	18,446,744,073,709,551,616

Conclusion

The given IPv6 address compresses to 2001:db8::ff00:42:8329 and expands back to its original notation. With a /64 prefix, the network prefix is 2001:db8:0:0::/64, leaving 64 bits for interface identifiers and providing 2^64 possible values. Unlike traditional IPv4 subnet calculations, IPv6 does not generally subtract two addresses for network and broadcast addresses.